

FUNDAMENTALS OF BIOLOGY LAB MANUAL



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Licensing

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About this Course

Hello, citizen scientist! You may be wondering why you're being called a "citizen scientist" or what that even means. This is an introductory biology course, after all. You may be taking the class as a prerequisite for nursing or maybe you are fulfilling your general education units as you work towards a degree in another field. Regardless of what your long term educational and career goals are, it is important not only to develop a general understanding of science, but also to recognize that science is happening all around us and that science affects our lives whether we realize it or not. It is in our best interest to understand science so that we can make well-informed, good decisions for ourselves, those around us, and the world at large.

Enter the citizen scientist. A citizen scientist is just that -- a regular person without an extensive formal or professional background in science who not only understands the importance of science but also participates in the vast scientific work happening around the globe. While professional scientists may hold impressive degrees and conduct cutting-edge research, sometimes they need the help of everyday people -- our help.

What do you need to be a citizen scientist? Not much, really. You only need to understand the scientific process, have a basic understanding of core concepts, have an openness to scientific evidence, and an ability to tap into your curiosity.

We will begin our journey as citizen scientists in this course by developing a good foundation in the basics of biology. To accomplish this, we will take on the role of scientists and perform thoughtful, hands-on experiments in the hopes that we will develop a respect for the scientific process and better understand how life all around us works. And hopefully, this course will spur you to join the citizen science community going into the future too.

Safety in the Laboratory

Science can be exciting and fun, but there are also many dangers in the laboratory. Many experiments require the use of harmful chemicals, flame, or sharp objects. While there's no need to fear these things, it's important to understand the dangers found in a lab classroom and what precautions we can take to prevent mishaps. You'll find that lab classrooms have several basic safety rules that you must follow to keep everyone safe. To ensure that you fully understand these rules, you will sign a safety contract included in this manual. Only sign the safety contract after the instructor has reviewed the rules with the class and you have asked for clarification on anything you did not understand. If you violate these lab rules, you may be asked to leave the lab. Remember, we are all in charge of safety!

1. Do not eat or drink in the laboratory. Any food and drink should be sealed and/or stored in a location deemed appropriate by the instructor and/or campus safety protocol.
2. Dress in lab-appropriate attire. Lab-appropriate attire generally means closed toed shoes and pants (or equivalent) covering most of the leg. Keep long hair tied back and confine or remove loose jewelry. If a lab requires more strict attire policies, abide by those.
3. Keep items such as bags and backpacks under the table instead of in walking spaces and on table tops to reduce trip hazards and contamination.
4. Use protective gear when directed. Some labs may require skin or eye protection. If the instructor and/or protocol require the use of gloves or goggles, remember that it is not optional.
5. Inform your instructor if you have medical concerns or situations that may require special precautions, such as allergies, pregnancy, or immunosuppression.
6. Identify and follow special safety precautions. When special attention is needed, your lab manual will utilize special symbols to grab your attention.
 - a.  The biohazard symbol indicates that a procedure may pose health concerns.
 - b.  The caution symbol indicates that special attention to safety should be given and identifies hazards that may come from substances and/or procedures.
7. Dispose of materials in their appropriate receptacle. Some items may be washed down the drain or thrown in the regular trash, but other items may require collection for special disposal such as certain chemical waste products, biohazardous waste, and sharp instruments. If you don't know where to dispose of something, always ask your instructor.
8. Report all spills, glass breakages, and accidents to the instructor. Do not attempt to clean any spillage, broken glass, or wounds yourself without direct instruction.
9. If a chemical or microorganism gets on your skin, wash the area immediately and thoroughly with mild soap and water.
10. Know the locations of safety equipment in the classroom. Safety equipment includes items like the telephone, emergency exits, fire extinguisher, fire blanket, eyewash station, emergency shower, first-aid kit, spill kit, and broken glass container.
11. Handle laboratory equipment with care. Many of the tools and models used in the lab are used by many classes and may not be easily replaced.
12. Pay special caution when using flame or heat. Never grab suspected or known hot objects with your hands and never leave a flame or hot plate unattended.
13. Report any unsafe actions or situations to the instructor, to include things like broken equipment, horseplay among students, or unauthorized guests.
14. Clean your workspace after completing your lab. This includes putting away equipment, sanitizing the table top, and disposing of any trash. Wash your hands before leaving the classroom.
15. Lastly, follow all other safety instructions given by the instructor that may not be covered in this form. Safety can be specific to the procedure being conducted and further safety measures may be required.

I, _____, fully understand and agree to adhere to the safety policies as explained in this safety contract. I understand that failure to adhere to these rules may result in me being asked to remedy the violation (when possible) or leave the lab classroom and that repeated violations may result in further penalties.

_____ Signature Date